<u>Quantitative Time-Series Analysis: US Sector ETFs</u>

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Dataset and Source

This project analyzes daily price data for five U.S. equity ETFs, SPY (S&P 500 benchmark), XLK (Technology), XLF (Financials), XLE (Energy), and XLV (Healthcare). I picked these four sectors because they are affected by different economic factors. Technology tracks growth, financials tracks rates, energy tracks commodities, and healthcare is defensive. Therefore, this gives many unique behaviors to compare across the sectors.

Data was pulled from Yahoo Finance. The timeline of the dataset is January 2, 2015 through April 17, 2026. This marks 2839 trading days in total. All prices are adjusted for splits and dividends. To ensure data was pulled correctly, I cross-checked values against Yahoo's website on the March 3rd, 2020 COVID bottom.

Analysis Methods

- **Preprocessing:** Converted adjusted-close prices to daily log returns. These are additive over time and closer to normally distributed than simple returns. Computed annualized return, volatility, and Sharpe ratio for each ticker.
- **Exploratory analysis:** Produced cumulative return curves, 30-day annualized volatility, a pairwise correlation heatmap, rolling 60-day correlations of each sector with SPY, and ACF/PACF plots of SPY returns.
- **Modeling:** Ran the Augmented Dickey-Fuller test to confirm the return series is stationary, then grid searched ARIMA (p, 0, q) for p, q in {0, 1, 2}, ranking candidates by AIC. Fit the winning model on the first 80% of the data and evaluate it on the remaining 20% using root mean squared error and mean absolute error against a naive zero predicted baseline.
- **Diagnostics:** Applied Ljun-Box, Jarque-Bera, and heteroskedasticity tests on the residuals to check what the model did and did not capture.

Key Findings

1. SPY delivered a higher Sharpe ratio, return per unit of risk, than Financials, Healthcare, and Energy, and only slightly trailed Technology. Energy was clearly behind as it produced less than 7% annualized return for nearly 30% annualized volatility and left investors underwater for roughly seven years. Broad-market exposure captured most of the upside with visibly smoother drawdowns
2. Sector correlations tightened during crises. In normal conditions, rolling correlations between sectors and the market sat in the 0.75-0.90 range, but during the March 2020 COVID crash and the 2022 drawdown they spiked toward 0.95 and above, Sectors stopped moving by themselves and started moving together. This means diversification benefits shrank in the moments they mattered most.
3. The forecasting model barely beat a predicted zero baseline. The chosen time-series model produced an average daily error of 0.0101, compared to 0.0102 for a baseline that predicts zero return every day, only a sub-1% improvement. This is consistent with the weak-form efficient hypothesis: the idea that past prices alone cannot meaningfully predict future prices. This sets realistic expectations for any analysis using only the market's own history.
4. The real predictability in market data is found in volatility, not in returns. Diagnostic checks confirmed the forecast captured every part of the linear pattern in the return series, but they also revealed two structural features no linear model can handle: extreme daily moves occur far more often than a normal bell curve predicts, and volatility itself shifts dramatically over time. These two features, fat tails and volatility clustering, are what actually drive market risk, and a volatility-focused model would be the next step in analysis.

Visualizations

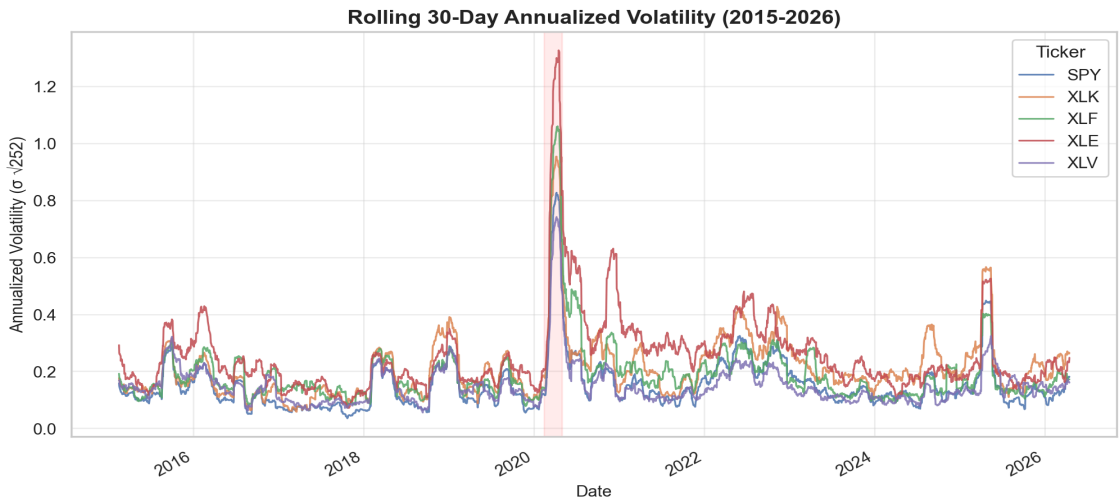

Figure 1. Rolling 30-Day Annualized Volatility (2015-2026)

Volatility shows sharp clustering across all five tickers. Energy (XLE) peaked around 130% annualized during the March 2020 COVID crash, almost 8x its typical volatility level. Vol spikes happen across every sector simultaneously which is consistent with a systemic factor that dominates sector-specific behavior during market stress.

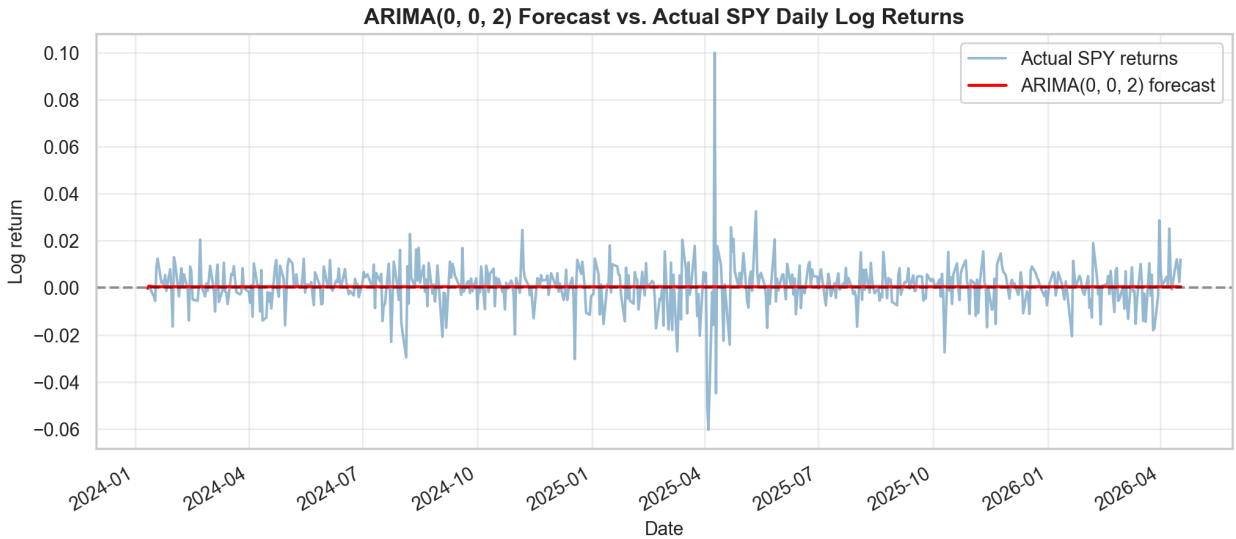

Figure 2. ARIMA (0, 0, 2) Forecast vs. Actual SPY Daily Log Returns

The out-of-sample forecast (red) hovers near zero while the actual reruns (blue) scatter wildly around it. This is a visualization of weak-form market efficiency at the daily horizon as past SPY returns contain only trace predictability for future returns.

Reflection

This project helped me build a full quantitative analysis, taking me through the work flow of pulling and cleaning real market data, visually exploring it, fitting and model, and finally interpreting the model's capabilities. One major lesson I learned was that a model failing to predict a certain outcome can be just as informative as one that succeeds. In this case, the near-zero forecasting edge was not a useless data point but a signal pointing toward where the real predictability in market data actually lives.

The exploratory work produced its own useful findings. The broad market outperformed most individual sectors on a risk-adjusted basis over this period, and correlations between sectors tightened significantly during crises, a pattern the static correlation view alone would miss. Together, these observations reinforce how much insight can come from straightforward visualizations before any modeling is involved.

Despite the great findings from this project, the analysis has major limits that are worth pointing out. The dataset only covers one major systemic crisis, COVID, which likely is a driving force for shapes of some of the patterns I observed. Additionally, the model I used is built for predicting averages rather than the volatility swings that drive most real-world market risk.

A next step in the project that I see is worth implementing would be a model designed specifically for volatility rather than returns, along with bringing in external signals, like interest rates or a broad market fear index, since past prices alone clearly don't give us the full picture. Overall, this project helped me realize that finding a model's incapabilities is just as valuable and can point us in the right direction for future work.